

Airbnb Clone

1. Project Overview

The Airbnb Clone is a full-stack vacation-rental marketplace inspired by the Airbnb listing experience. The project focuses on creating a responsive, pixel-accurate interface with dynamic search, property data, interactive components, and backend API integration.

The frontend communicates with a Node.js/Express backend, while MongoDB is used to store and manage application data.

2. Technology Stack

Frontend

- React 19
- TypeScript
- Vite
- Tailwind CSS v4
- React Router
- TanStack Query
- Lucide React

Backend

- Node.js
- Express.js
- TypeScript
- MongoDB
- Mongoose

3. Main Features

- Airbnb-style responsive UI
- Dynamic destination search
- Date selection
- Property/listing data from backend
- Interactive search bar
- Responsive design for desktop and mobile
- Smooth dropdowns, modals, and animations
- REST API integration
- MongoDB-based dynamic data
- Realistic seeded property, location, host, and review data

4. Project Architecture

The application follows a **Client - Server Architecture**.

Frontend → REST API → Backend → MongoDB

5. Production Scaling Approach

For production-scale deployment, the application can be extended with:

- **CDN:** Cloudflare / AWS CloudFront
- **Backend Scaling:** Docker + Kubernetes
- **Caching:** Redis
- **Search:** Elasticsearch / Algolia
- **Storage:** AWS S3 / Cloudinary
- **Database:** MongoDB with scaling or PostgreSQL for transactional data
- **CI/CD:** GitHub Actions
- **Infrastructure:** Terraform
- **API Management:** API Gateway

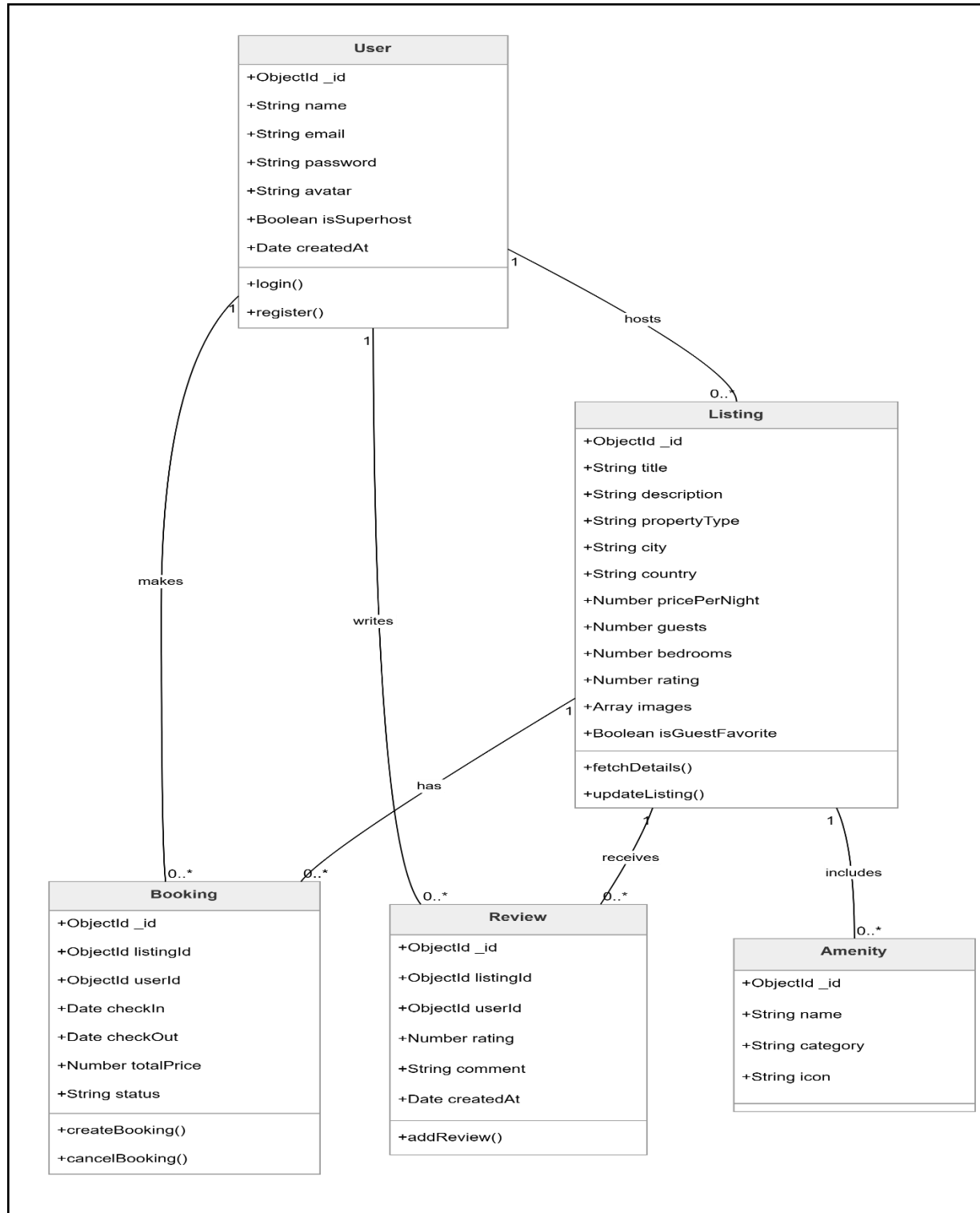
6. AI-Assisted Development

AI development tools were used to assist with project scaffolding, debugging, refactoring, Tailwind CSS configuration, API development, database seeding, and improving UI interactions.

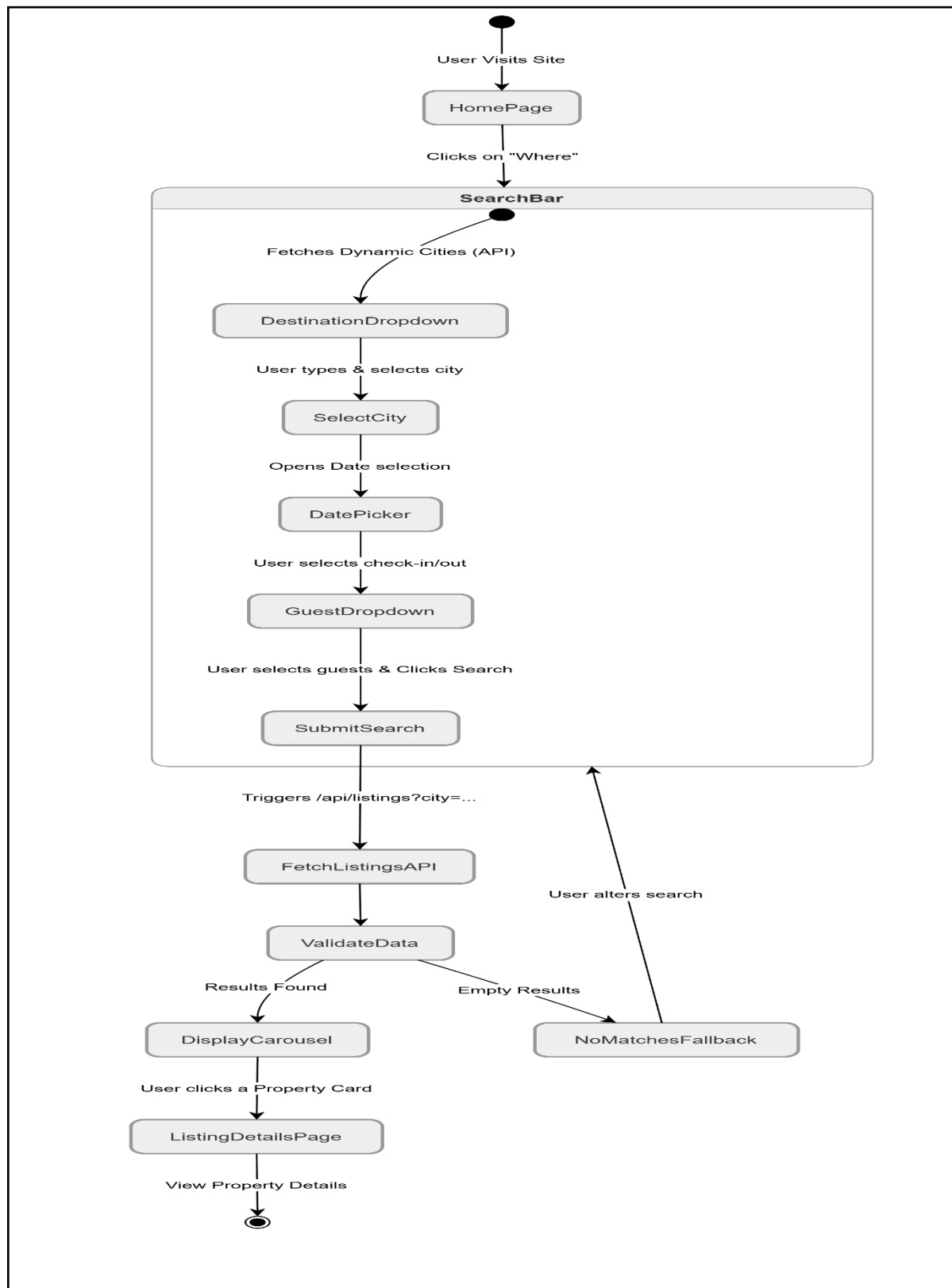
7. Conclusion

The project demonstrates a complete full-stack vacation-rental marketplace with a modern React frontend, RESTful backend, MongoDB database, responsive UI, dynamic data, and a scalable architecture designed for future production deployment.

8. Class Diagram



9. Activity Diagram



10. Data Flow Diagram

